

Appendix G - Coding Agents

Vibe Coding: A Starting Point

"Vibe coding" has become a powerful technique for rapid innovation and creative exploration. This practice involves using LLMs to generate initial drafts, outline complex logic, or build quick prototypes, significantly reducing initial friction. It is invaluable for overcoming the "blank page" problem, enabling developers to quickly transition from a vague concept to tangible, runnable code. Vibe coding is particularly effective when exploring unfamiliar APIs or testing novel architectural patterns, as it bypasses the immediate need for perfect implementation. The generated code often acts as a creative catalyst, providing a foundation for developers to critique, refactor, and expand upon. Its primary strength lies in its ability to accelerate the initial discovery and ideation phases of the software lifecycle. However, while vibe coding excels at brainstorming, developing robust, scalable, and maintainable software demands a more structured approach, shifting from pure generation to a collaborative partnership with specialized coding agents.

Agents as Team Members

While the initial wave focused on raw code generation—the "vibe code" perfect for ideation—the industry is now shifting towards a more integrated and powerful paradigm for production work. The most effective development teams are not merely delegating tasks to Agent; they are augmenting themselves with a suite of sophisticated coding agents. These agents act as tireless, specialized team members, amplifying human creativity and dramatically increasing a team's scalability and velocity.

This evolution is reflected in statements from industry leaders. In early 2025, Alphabet CEO Sundar Pichai noted that at Google, *"over 30% of new code is now assisted or generated by our Gemini models, fundamentally changing our development velocity."* Microsoft made a similar claim. This industry-wide shift signals that the true frontier is not replacing developers, but empowering them. The goal is an augmented relationship where humans guide the architectural vision and creative problem-solving, while agents handle specialized, scalable tasks like testing, documentation, and review.

This chapter presents a framework for organizing a human-agent team based on the core philosophy that human developers act as creative leads and architects, while AI agents function as force multipliers. This framework rests upon three foundational principles:

1. **Human-Led Orchestration:** The developer is the team lead and project architect. They are always in the loop, orchestrating the workflow, setting the high-level goals, and making the final decisions. The agents are powerful, but they are supportive collaborators. The developer directs which agent to engage, provides the necessary

context, and, most importantly, exercises the final judgment on any Agent-generated output, ensuring it aligns with the project's quality standards and long-term vision.

2. **The Primacy of Context:** An agent's performance is entirely dependent on the quality and completeness of its context. A powerful LLM with poor context is useless. Therefore, our framework prioritizes a meticulous, human-led approach to context curation. Automated, black-box context retrieval is avoided. The developer is responsible for assembling the perfect "briefing" for their Agent team member. This includes:
 - **The Complete Codebase:** Providing all relevant source code so the agent understands the existing patterns and logic.
 - **External Knowledge:** Supplying specific documentation, API definitions, or design documents.
 - **The Human Brief:** Articulating clear goals, requirements, pull request descriptions, and style guides.
3. **Direct Model Access:** To achieve state-of-the-art results, the agents must be powered by direct access to frontier models (e.g., Gemini 2.5 PRO, Claude Opus 4, OpenAI, DeepSeek, etc). Using less powerful models or routing requests through intermediary platforms that obscure or truncate context will degrade performance. The framework is built on creating the purest possible dialogue between the human lead and the raw capabilities of the underlying model, ensuring each agent operates at its peak potential.

The framework is structured as a team of specialized agents, each designed for a core function in the development lifecycle. The human developer acts as the central orchestrator, delegating tasks and integrating the results.

Core Components

To effectively leverage a frontier Large Language Model, this framework assigns distinct development roles to a team of specialized agents. These agents are not separate applications but are conceptual personas invoked within the LLM through carefully crafted, role-specific prompts and contexts. This approach ensures that the model's vast capabilities are precisely focused on the task at hand, from writing initial code to performing a nuanced, critical review.

The Orchestrator: The Human Developer: In this collaborative framework, the human developer acts as the Orchestrator, serving as the central intelligence and ultimate authority over the AI agents.

- **Role:** Team Lead, Architect, and final decision-maker. The orchestrator defines tasks, prepares the context, and validates all work done by the agents.
- **Interface:** The developer's own terminal, editor, and the native web UI of the chosen Agents.

The Context Staging Area: As the foundation for any successful agent interaction, the Context Staging Area is where the human developer meticulously prepares a complete and task-specific briefing.

- **Role:** A dedicated workspace for each task, ensuring agents receive a complete and accurate briefing.
- **Implementation:** A temporary directory (task-context/) containing markdown files for goals, code files, and relevant docs

The Specialist Agents: By using targeted prompts, we can build a team of specialist agents, each tailored for a specific development task.

- **The Scaffolder Agent: The Implementer**
 - **Purpose:** Writes new code, implements features, or creates boilerplate based on detailed specifications.
 - **Invocation Prompt:** "You are a senior software engineer. Based on the requirements in O1_BRIEF.md and the existing patterns in O2_CODE/, implement the feature..."
- **The Test Engineer Agent: The Quality Guard**
 - **Purpose:** Writes comprehensive unit tests, integration tests, and end-to-end tests for new or existing code.
 - **Invocation Prompt:** "You are a quality assurance engineer. For the code provided in O2_CODE/, write a full suite of unit tests using [Testing Framework, e.g., pytest]. Cover all edge cases and adhere to the project's testing philosophy."
- **The Documenter Agent: The Scribe**
 - **Purpose:** Generates clear, concise documentation for functions, classes, APIs, or entire codebases.
 - **Invocation Prompt:** "You are a technical writer. Generate markdown documentation for the API endpoints defined in the provided code. Include request/response examples and explain each parameter."
- **The Optimizer Agent: The Refactoring Partner**
 - **Purpose:** Proposes performance optimizations and code refactoring to improve readability, maintainability, and efficiency.
 - **Invocation Prompt:** "Analyze the provided code for performance bottlenecks or areas that could be refactored for clarity. Propose specific changes with explanations for why they are an improvement."
- **The Process Agent: The Code Supervisor**
 - **Critique:** The agent performs an initial pass, identifying potential bugs, style violations, and logical flaws, much like a static analysis tool.
 - **Reflection:** The agent then analyzes its own critique. It synthesizes the findings, prioritizes the most critical issues, dismisses pedantic or low-impact suggestions, and provides a high-level, actionable summary for the human developer.

- **Invocation Prompt:** "You are a principal engineer conducting a code review. First, perform a detailed critique of the changes. Second, reflect on your critique to provide a concise, prioritized summary of the most important feedback."

Ultimately, this human-led model creates a powerful synergy between the developer's strategic direction and the agents' tactical execution. As a result, developers can transcend routine tasks, focusing their expertise on the creative and architectural challenges that deliver the most value.

Practical Implementation

Setup Checklist

To effectively implement the human-agent team framework, the following setup is recommended, focusing on maintaining control while improving efficiency.

1. **Provision Access to Frontier Models** Secure API keys for at least two leading large language models, such as Gemini 2.5 Pro and Claude 4 Opus. This dual-provider approach allows for comparative analysis and hedges against single-platform limitations or downtime. These credentials should be managed securely as you would any other production secret.
2. **Implement a Local Context Orchestrator** Instead of ad-hoc scripts, use a lightweight CLI tool or a local agent runner to manage context. These tools should allow you to define a simple configuration file (e.g., `context.toml`) in your project root that specifies which files, directories, or even URLs to compile into a single payload for the LLM prompt. This ensures you retain full, transparent control over what the model sees on every request.
3. **Establish a Version-Controlled Prompt Library** Create a dedicated `/prompts` directory within your project's Git repository. In it, store the invocation prompts for each specialist agent (e.g., `reviewer.md`, `documenter.md`, `tester.md`) as markdown files. Treating your prompts as code allows the entire team to collaborate on, refine, and version the instructions given to your AI agents over time.
4. **Integrate Agent Workflows with Git Hooks** Automate your review rhythm by using local Git hooks. For instance, a pre-commit hook can be configured to automatically trigger the Reviewer Agent on your staged changes. The agent's critique-and-reflection summary can be presented directly in your terminal, providing immediate feedback before you finalize the commit and baking the quality assurance step directly into your development process.

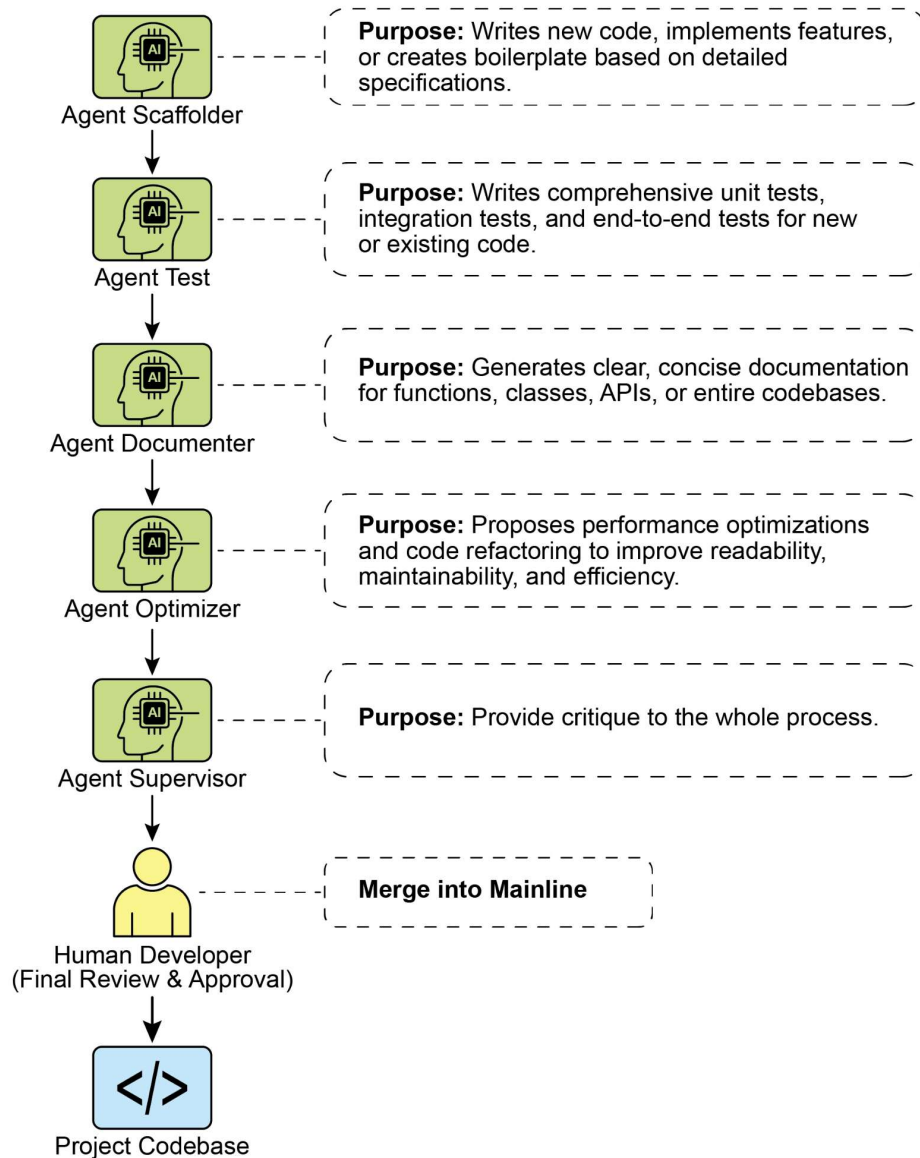


Fig. 1: Coding Specialist Examples

Principles for Leading the Augmented Team

Successfully leading this framework requires evolving from a sole contributor into the lead of a human-AI team, guided by the following principles:

- Maintain Architectural Ownership** Your role is to set the strategic direction and own the high-level architecture. You define the "what" and the "why," using the agent team to accelerate the "how." You are the final **arbiter** of design, ensuring every component aligns with the project's long-term vision and quality standards.

- **Master the Art of the Brief** The quality of an agent's output is a direct reflection of the quality of its input. Master the art of the brief by providing clear, unambiguous, and comprehensive context for every task. Think of your prompt not as a simple command, but as a complete briefing package for a new, highly capable team member.
- **Act as the Ultimate Quality Gate** An agent's output is always a proposal, never a command. Treat the Reviewer Agent's feedback as a powerful signal, but you are the ultimate quality gate. Apply your domain expertise and project-specific knowledge to validate, challenge, and approve all changes, acting as the final guardian of the codebase's integrity.
- **Engage in Iterative Dialogue** The best results emerge from conversation, not monologue. If an agent's initial output is imperfect, don't discard it—refine it. Provide corrective feedback, add clarifying context, and prompt for another attempt. This iterative dialogue is crucial, especially with the Reviewer Agent, whose "Reflection" output is designed to be the start of a collaborative discussion, not just a final report.

Conclusion

The future of code development has arrived, and it is augmented. The era of the lone coder has given way to a new paradigm where developers lead teams of specialized AI agents. This model doesn't diminish the human role; it elevates it by automating routine tasks, scaling individual impact, and achieving a development velocity previously unimaginable.

By offloading tactical execution to Agents, developers can now dedicate their cognitive energy to what truly matters: strategic innovation, resilient architectural design, and the creative problem-solving required to build products that delight users. The fundamental relationship has been redefined; it is no longer a contest of human versus machine, but a partnership between human ingenuity and AI, working as a single, seamlessly integrated team.

References

1. AI is responsible for generating more than 30% of the code at Google
https://www.reddit.com/r/singularity/comments/1k7rxo0/ai_is_now_writing_well_over_30_of_the_code_at/
2. AI is responsible for generating more than 30% of the code at Microsoft
<https://www.businesstoday.in/tech-today/news/story/30-of-microsofts-code-is-now-ai-generated-says-ceo-satya-nadella-474167-2025-04-30>

Conclusion

Throughout this book we have journeyed from the foundational concepts of agentic AI to the practical implementation of sophisticated, autonomous systems. We began with the premise that building intelligent agents is akin to creating a complex work of art on a technical canvas—a process that requires not just a powerful cognitive engine like a large language model, but also a robust set of architectural blueprints. These blueprints, or agentic patterns, provide the structure and reliability needed to transform simple, reactive models into proactive, goal-oriented entities capable of complex reasoning and action.

This concluding chapter will synthesize the core principles we have explored. We will first review the key agentic patterns, grouping them into a cohesive framework that underscores their collective importance. Next, we will examine how these individual patterns can be composed into more complex systems, creating a powerful synergy. Finally, we will look ahead to the future of agent development, exploring the emerging trends and challenges that will shape the next generation of intelligent systems.

Review of key agentic principles

The 21 patterns detailed in this guide represent a comprehensive toolkit for agent development. While each pattern addresses a specific design challenge, they can be understood collectively by grouping them into foundational categories that mirror the core competencies of an intelligent agent.

1. **Core Execution and Task Decomposition:** At the most fundamental level, agents must be able to execute tasks. The patterns of Prompt Chaining, Routing, Parallelization, and Planning form the bedrock of an agent's ability to act. Prompt Chaining provides a simple yet powerful method for breaking down a problem into a linear sequence of discrete steps, ensuring that the output of one operation logically informs the next. When workflows require more dynamic behavior, Routing introduces conditional logic, allowing an agent to select the most appropriate path or tool based on the context of the input. Parallelization optimizes efficiency by enabling the concurrent execution of independent sub-tasks, while the Planning pattern elevates the agent from a mere executor to a strategist, capable of formulating a multi-step plan to achieve a high-level objective.

2. **Interaction with the External Environment:** An agent's utility is significantly enhanced by its ability to interact with the world beyond its immediate internal state. The Tool Use (Function Calling) pattern is paramount here, providing the mechanism for agents to leverage external APIs, databases, and other software systems. This grounds the agent's operations in real-world data and capabilities. To effectively use these tools, agents must often access specific, relevant information from vast repositories. The Knowledge Retrieval pattern, particularly Retrieval-Augmented Generation (RAG), addresses this by enabling agents to query knowledge bases and incorporate that information into their responses, making them more accurate and contextually aware.
3. **State, Learning, and Self-Improvement:** For an agent to perform more than just single-turn tasks, it must possess the ability to maintain context and improve over time. The Memory Management pattern is crucial for endowing agents with both short-term conversational context and long-term knowledge retention. Beyond simple memory, truly intelligent agents exhibit the capacity for self-improvement. The Reflection and Self-Correction patterns enable an agent to critique its own output, identify errors or shortcomings, and iteratively refine its work, leading to a higher quality final result. The Learning and Adaptation pattern takes this a step further, allowing an agent's behavior to evolve based on feedback and experience, making it more effective over time.
4. **Collaboration and Communication:** Many complex problems are best solved through collaboration. The Multi-Agent Collaboration pattern allows for the creation of systems where multiple specialized agents, each with a distinct role and set of capabilities, work together to achieve a common goal. This division of labor enables the system to tackle multifaceted problems that would be intractable for a single agent. The effectiveness of such systems hinges on clear and efficient communication, a challenge addressed by the Inter-Agent Communication (A2A) and Model Context Protocol (MCP) patterns, which aim to standardize how agents and tools exchange information.

These principles, when applied through their respective patterns, provide a robust framework for building intelligent systems. They guide the developer in creating agents that are not only capable of performing complex tasks but are also structured, reliable, and adaptable.

Combining Patterns for Complex Systems

The true power of agentic design emerges not from the application of a single pattern in isolation, but from the artful composition of multiple patterns to create

sophisticated, multi-layered systems. The agentic canvas is rarely populated by a single, simple workflow; instead, it becomes a tapestry of interconnected patterns that work in concert to achieve a complex objective.

Consider the development of an autonomous AI research assistant, a task that requires a combination of planning, information retrieval, analysis, and synthesis. Such a system would be a prime example of pattern composition:

- **Initial Planning:** A user query, such as "Analyze the impact of quantum computing on the cybersecurity landscape," would first be received by a Planner agent. This agent would leverage the Planning pattern to decompose the high-level request into a structured, multi-step research plan. This plan might include steps like "Identify foundational concepts of quantum computing," "Research common cryptographic algorithms," "Find expert analyses on quantum threats to cryptography," and "Synthesize findings into a structured report."
- **Information Gathering with Tool Use:** To execute this plan, the agent would rely heavily on the Tool Use pattern. Each step of the plan would trigger a call to a Google Search or `vertex_ai_search` tool. For more structured data, it might use tools to query academic databases like ArXiv or financial data APIs.
- **Collaborative Analysis and Writing:** A single agent might handle this, but a more robust architecture would employ Multi-Agent Collaboration. A "Researcher" agent could be responsible for executing the search plan and gathering raw information. Its output—a collection of summaries and source links—would then be passed to a "Writer" agent. This specialist agent, using the initial plan as its outline, would synthesize the collected information into a coherent draft.
- **Iterative Reflection and Refinement:** A first draft is rarely perfect. The Reflection pattern could be implemented by introducing a third "Critic" agent. This agent's sole purpose would be to review the Writer's draft, checking for logical inconsistencies, factual inaccuracies, or areas lacking clarity. Its critique would be fed back to the Writer agent, which would then leverage the Self-Correction pattern to refine its output, incorporating the feedback to produce a higher-quality final report.
- **State Management:** Throughout this entire process, a Memory Management system would be essential. It would maintain the state of the research plan, store the information gathered by the Researcher, hold the drafts created by the Writer, and track the feedback from the Critic, ensuring that context is preserved across the entire multi-step, multi-agent workflow.

In this example, at least five distinct agentic patterns are woven together. The Planning pattern provides the high-level structure, Tool Use grounds the operation in real-world data, Multi-Agent Collaboration enables specialization and division of labor, Reflection ensures quality, and Memory Management maintains coherence. This composition transforms a set of individual capabilities into a powerful, autonomous system capable of tackling a task that would be far too complex for a single prompt or a simple chain.

Looking to the Future

The composition of agentic patterns into complex systems, as illustrated by our AI research assistant, is not the end of the story but rather the beginning of a new chapter in software development. As we look ahead, several emerging trends and challenges will define the next generation of intelligent systems, pushing the boundaries of what is possible and demanding even greater sophistication from their creators.

The journey toward more advanced agentic AI will be marked by a drive for greater **autonomy and reasoning**. The patterns we have discussed provide the scaffolding for goal-oriented behavior, but the future will require agents that can navigate ambiguity, perform abstract and causal reasoning, and even exhibit a degree of common sense. This will likely involve tighter integration with novel model architectures and neuro-symbolic approaches that blend the pattern-matching strengths of LLMs with the logical rigor of classical AI. We will see a shift from human-in-the-loop systems, where the agent is a co-pilot, to human-on-the-loop systems, where agents are trusted to execute complex, long-running tasks with minimal oversight, reporting back only when the objective is complete or a critical exception occurs.

This evolution will be accompanied by the rise of **agentic ecosystems and standardization**. The Multi-Agent Collaboration pattern highlights the power of specialized agents, and the future will see the emergence of open marketplaces and platforms where developers can deploy, discover, and orchestrate fleets of agents-as-a-service. For this to succeed, the principles behind the Model Context Protocol (MCP) and Inter-Agent Communication (A2A) will become paramount, leading to industry-wide standards for how agents, tools, and models exchange not just data, but also context, goals, and capabilities.

A prime example of this growing ecosystem is the "Awesome Agents" GitHub repository, a valuable resource that serves as a curated list of open-source AI agents, frameworks, and tools. It showcases the rapid innovation in the field by organizing cutting-edge projects for applications ranging from software development to autonomous research and conversational AI.

However, this path is not without its formidable challenges. The core issues of **safety, alignment, and robustness** will become even more critical as agents become more autonomous and interconnected. How do we ensure an agent's learning and adaptation do not cause it to drift from its original purpose? How do we build systems that are resilient to adversarial attacks and unpredictable real-world scenarios? Answering these questions will require a new set of "safety patterns" and a rigorous engineering discipline focused on testing, validation, and ethical alignment.

Final Thoughts

Throughout this guide, we have framed the construction of intelligent agents as an art form practiced on a technical canvas. These Agentic Design patterns are your palette and your brushstrokes—the foundational elements that allow you to move beyond simple prompts and create dynamic, responsive, and goal-oriented entities. They provide the architectural discipline needed to transform the raw cognitive power of a large language model into a reliable and purposeful system.

The true craft lies not in mastering a single pattern but in understanding their interplay—in seeing the canvas as a whole and composing a system where planning, tool use, reflection, and collaboration work in harmony. The principles of agentic design are the grammar of a new language of creation, one that allows us to instruct machines not just on what to do, but on how to *be*.

The field of agentic AI is one of the most exciting and rapidly evolving domains in technology. The concepts and patterns detailed here are not a final, static dogma but a starting point—a solid foundation upon which to build, experiment, and innovate. The future is not one where we are simply users of AI, but one where we are the architects of intelligent systems that will help us solve the world's most complex problems. The canvas is before you, the patterns are in your hands. Now, it is time to build.